

SANJAY KUMAR

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SUMMARY

Full Stack Software Engineer with 2+ years of experience building secure web applications and integrating AI-powered capabilities into production systems. Experienced in React, JavaScript/TypeScript, C#, .NET, Node.js, REST APIs, SQL Server, MongoDB, automated testing, browser security, and performance optimization. Hands-on experience evaluating AI approaches and taking proof-of-concept solutions through production integration.

SKILLS

- **Programming Languages:** JavaScript, TypeScript, C#, SQL
- **Frontend Development:** React.js, HTML5, CSS3, jQuery
- **Backend Development:** C#, .NET, .NET Core, .NET Framework, Node.js, Express.js, REST APIs
- **AI & Computer Vision:** AI Model Integration, Face Verification, Identity & Document Verification
- **Databases:** SQL Server, MySQL, MongoDB
- **Testing & DevOps:** Mocha, Chai, xUnit, Azure Pipelines, CI/CD
- **Browser & Web:** Chrome Extension Development, Web Workers, Browser APIs

WORK EXPERIENCE

Internet Testing System, India

Full Stack Software Engineer • Coimbatore, India • Jan 2024 - Present

- Developed secure web-based examination and candidate verification applications supporting online testing and remote proctoring workflows.
- Integrated AI-powered capabilities for candidate face verification, identity/document verification, and examination-environment checks from proof of concept through production.
- Researched and evaluated AI approaches based on accuracy, response time, resource usage, and performance requirements; helped transition inference workloads toward server-side processing.
- Used Web Workers and background processing to reduce browser main-thread load and improve application responsiveness while integrating AI functionality into an existing production system.
- Developed a Chrome extension for controlled examination environments, restricting tabs, URLs, copy/paste, keyboard shortcuts, developer tools, screenshots, and navigation.
- Built backend functionality using C#, .NET/.NET Core, Node.js, and REST APIs, including authentication and authorization for controlled access to application resources.
- Refactored and optimized legacy production code without disrupting existing business workflows; identified and addressed application-level security vulnerabilities.
- Developed automated tests with Mocha/Chai and xUnit, integrating test execution into Azure Pipelines for pull-request validation.

PSG Institute of Management, Coimbatore, India

Full Stack Web Developer Intern • Aug 2022 - Dec 2022

- Developed an alumni webpage as part of a web development internship, contributing to the implementation of the application interface and functionality.
- Worked on frontend and backend components and supported the development of a structured web experience for alumni information and engagement.

PROJECTS

- **AI Object Detection & Face Recognition Web Application** - Built a real-time web application using live camera feeds for object detection and face recognition; implemented facial comparison using facial embeddings to support identity verification.
- **Alumni Networking Platform** - Built a full-featured alumni networking platform with blog posts, job boards, news feeds, administrative content management, user profiles, and supporting frontend/backend database structures.

EDUCATION

Master of Science in Software Systems • PSG College of Arts and Science, Coimbatore, India • 2019 - 2024

CERTIFICATIONS

• **GitHub Copilot Certification (GH-300)** • Azure AI Fundamentals • AWS Certified Cloud Practitioner • CS50: Introduction to Computer Science (Harvard University) • Azure Data Fundamentals